

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE																																
Supplementary Winter-2023																																
Course: B. Tech. Branch :Computer Engineering and Allied Semester :IV																																
Subject Code & Name: BTBSC404 Probability &Statistics																																
Max Marks: 60 Date:23/01/24 Duration: 3 Hr.																																
Instructions to the Students:																																
1. All the questions are compulsory.																																
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.																																
3. Use of non-programmable scientific calculators is allowed.																																
4. Assume suitable data wherever necessary and mention it clearly.																																
								(Level /CO)	Marks																							
Q. 1	Solve Any Two of the following.									12																						
A)	There are 3 true coins and 1 false coin with 'head' on both sides. A coin is chosen at random and tossed 4 times. If 'head' occurs all the 4 times, what is the probability that the false coin has been chosen and used?								CO-2	6																						
B)	If A and B are any 2 events such that $P(A) = \frac{3}{4}$ and $P(B) = \frac{5}{8}$. Prove that $\frac{3}{8} \leq P(A \cap B) \leq \frac{5}{8}$								CO-2	6																						
C)	A box contains 5 green pencils and 7 yellow pencils. Two pencils are chosen at random from the box without replacement. What is the probability that both are yellow?								CO-2	6																						
Q.2	Solve Any Two of the following.									12																						
A)	Find the value of k , if the following function is a probability density function. $f(x) = \begin{cases} k(x-1)^3 & 1 \leq x \leq 3 \\ 0 & otherwise \end{cases}$								CO-1	6																						
B)	A firm has two cars which it hires out day by day. The number of demands of a car each day is distributed as a Poisson variate of mean 1.5. Calculate the probable number of days in a year on which (i) neither car is in demand (ii) a demand is refused.								CO-1	6																						
C)	Five coins are tossed 100 times and the following results were obtained <table border="1"><tr><td>No. of heads</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>frequency</td><td>10</td><td>20</td><td>30</td><td>15</td><td>15</td><td>10</td></tr></table> Fit a Binomial distribution.								No. of heads	0	1	2	3	4	5	frequency	10	20	30	15	15	10	CO-1	6								
No. of heads	0	1	2	3	4	5																										
frequency	10	20	30	15	15	10																										
Q. 3	Solve Any Two of the following.									12																						
A)	Calculate Karl Pearson's co-efficients of correlation from the following data <table border="1"><tr><td>x</td><td>28</td><td>45</td><td>40</td><td>38</td><td>35</td><td>33</td><td>40</td><td>32</td><td>36</td><td>33</td></tr><tr><td>y</td><td>23</td><td>34</td><td>33</td><td>34</td><td>30</td><td>26</td><td>28</td><td>31</td><td>36</td><td>35</td></tr></table>								x	28	45	40	38	35	33	40	32	36	33	y	23	34	33	34	30	26	28	31	36	35	CO-3	6
x	28	45	40	38	35	33	40	32	36	33																						
y	23	34	33	34	30	26	28	31	36	35																						

B)	Calculate the value of rank correlation coefficient from the following data regarding marks of six students in statistics and accountancy in a test	CO-4	6																		
	<table><tr><td>Marks in statistics</td><td>40</td><td>42</td><td>45</td><td>35</td><td>36</td><td>39</td></tr><tr><td>Marks in accountancy</td><td>46</td><td>43</td><td>44</td><td>39</td><td>40</td><td>43</td></tr></table>	Marks in statistics	40	42	45	35	36	39	Marks in accountancy	46	43	44	39	40	43						
Marks in statistics	40	42	45	35	36	39															
Marks in accountancy	46	43	44	39	40	43															
C)	Prove that limits of correlation coefficients are lies between $-1 \leq r \leq 1$.	CO-4	6																		
Q.4	Solve Any Two of the following.		12																		
A)	Find the equations of the lines of regression on (i) Y on X and (ii) Xon Y and also a co-efficient of correlation from the following table.	CO-3	6																		
	<table><tr><td>X</td><td>62</td><td>64</td><td>65</td><td>69</td><td>70</td><td>71</td><td>72</td><td>74</td></tr><tr><td>Y</td><td>126</td><td>125</td><td>139</td><td>145</td><td>165</td><td>152</td><td>180</td><td>208</td></tr></table>	X	62	64	65	69	70	71	72	74	Y	126	125	139	145	165	152	180	208		
X	62	64	65	69	70	71	72	74													
Y	126	125	139	145	165	152	180	208													
B)	The equations to the two lines of regression are $6y = 5x + 90$ and $15x = 8y + 130$. Find (i) the means of x and y , (ii) the coefficient of correlation ,(iii) if variance of $x = 16$, find also the standard deviation of y .	CO-3	6																		
C)	If the coefficient of correlation between two variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}\left(\frac{3}{5}\right)$. Prove that $\sigma_x = \frac{1}{2}\sigma_y$.	CO-3	6																		
Q. 5	Solve Any Two of the following.		12																		
A)	A manufacturer claims that only 4% of his products supplied by him are defective. A random sample of 600 products contained 36 defectives. Test the claim of the manufacturer.	CO-4	6																		
B)	A machine produced 16 defectives articles in a batch of 500. After overhauling it produced 3 defectives in a batch of 100. Has the machine improved?	CO-4	6																		
C)	A soap manufacturing company was distributing a particular brand of soap through a large number of retail soap. Before a heavy advertisement campaign, the mean sale per week per shop was 140 dozens. After the campaign a sample of 26 shops was taken and the mean sale was found to be 147 dozens with standard deviation of 16. Can you consider the advertisement effective?	CO-5	6																		
	*** End ***																				